# Vector-Space Optimization for Contraction Theory-Based Control Design: An Energy-Based Effective Space Approach

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***Abstract*—Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.**

***Index Terms*—Contraction theory, optimization-based control, energy-based control**

## NOTATION

In this study, the following notation is used:

- $\mathbb{R}^n$ denotes the $n$-dimensional Euclidean space.
- $\mathbb{R}^{n\times m}$ denotes the set of $n \times m$ real matrices.
- $\otimes$ denotes the Kronecker product [1, Chap. 7 Def. 7.1.2].
- A vector and a matrix are denoted by $\boldsymbol{x} = [x_i]_{i\in\{1,\cdots,n\}} \in \mathbb{R}^n$ and $\boldsymbol{A} := [a_{ij}]_{i\in\{1,\cdots,n\},j\in\{1,\cdots,m\}} \in \mathbb{R}^{n\times m}$, respectively.
- $\mathrm{row}_i(\boldsymbol{A})$ denotes the $i^{\text{th}}$ row of the matrix $\boldsymbol{A} \in \mathbb{R}^{n\times m}$.
- For $\boldsymbol{A} \in \mathbb{R}^{n\times m}$, denotes the vectorization of $\boldsymbol{A}$, $\mathrm{vec}(\boldsymbol{A}) := (\mathrm{row}_1(\boldsymbol{A}^\top), \cdots, \mathrm{row}_m(\boldsymbol{A}^\top))^\top \in \mathbb{R}^{nm}$.
- $\lambda_{\min}(\boldsymbol{A})$ denotes the minimum eigenvalue of the matrix $\boldsymbol{A} \in \mathbb{R}^{n\times n}$.
- $\boldsymbol{I}_n$ denotes the $n \times n$ identity matrix, and $\boldsymbol{0}_{n\times m}$ denotes the $n \times m$ zero matrix.

## I. INTRODUCTION

### A. Motivation

### B. Related Works

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## II. PRELIMINARIES

## III. PROBLEM FORMULATION

## IV. MAIN RESULTS

### A. Projection of Contraction Condition

In this section, we present the proposed vector-space optimization framework. We project the contraction condition (??) onto the trajectory error vector space by pre- and post-multiplying it by $\boldsymbol{e}^\top$ and $\boldsymbol{e}$, respectively, as follows:

$$-2\boldsymbol{y}^\top \left(\boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^\top\right)\boldsymbol{y} + \boldsymbol{e}^\top\left(\tfrac{1}{T_s} + 2\alpha + 2\mathbb{A}^\top\right)\boldsymbol{y} - \tfrac{\boldsymbol{e}^\top\boldsymbol{M}_{\text{pre}}\boldsymbol{e}}{T_s} \le 0, \tag{1}$$

where $\boldsymbol{y} := \boldsymbol{M}\boldsymbol{e}$ denotes a metric-weighted error vector, serving as a new optimization variable. For discrete-time implementation, $\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{M}$ is approximated as $\frac{1}{T_s}(\boldsymbol{M} - \boldsymbol{M}_{\text{pre}})$, where $T_s$ denotes the sampling time and $\boldsymbol{M}_{\text{pre}}$ denotes the contraction metric obtained at the previous time step.

To justify the proposed approach, we first present the following theorem that provides a sufficient condition of the projected contraction condition (1) to guarantee the full contraction condition (??).

**Theorem 1.** *For the system (??) satisfying Assumption ?? with the control law (??), the projected contraction condition (1) is necessary for the full contraction condition (??). Furthermore, suppose that there exists a metric-weighted error vector $\boldsymbol{y} = \boldsymbol{M}\boldsymbol{e}$ satisfying (1). If the reconstructed matrix $\boldsymbol{M}$ from $\boldsymbol{y}$ has a sufficiently large error-directional metric scale $\mu$ and a sufficiently small condition number $\chi$, then the reconstructed matrix also satisfies the full contraction condition (??).*

*Proof. Necessity.*

If the full contraction condition (??) holds, then pre- and post-multiplying it by $\boldsymbol{e}^\top$ and $\boldsymbol{e}$ directly yields the projected condition (1).

Therefore, the projected condition is necessary for the full contraction condition.

*Sufficiency.*

Since the control effectiveness matrix $\boldsymbol{B}$ is full row rank matrix according to Assumption ??, we have

$$\boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^\top \ge \sigma\boldsymbol{I}_n, \tag{2}$$

for some $\sigma \in \mathbb{R}_{>0}$. In addition, according to Assumption ??, the SDC $\mathbb{A}$ is bounded such that $\|\mathbb{A}\| \le \overline{A}$, for some $\overline{A} \in \mathbb{R}_{>0}$.

The error-directional metric scale $\mu$ can be defined as follows:

$$\mu := \frac{\boldsymbol{e}^\top\boldsymbol{M}\boldsymbol{e}}{\boldsymbol{e}^\top\boldsymbol{e}}. \tag{3}$$

By the definition of $\boldsymbol{M}$, the following inequality holds:

$$\underline{m} \le \mu \le \overline{m} \implies \tfrac{\mu}{\chi} \le \underline{m},\ \overline{m} \le \mu\chi,\ 1 \le \chi. \tag{4}$$

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Each term of the projected contraction condition (1) can be conservatively upper-bounded as follows:

$$-2\|\boldsymbol{y}^\top\left(\boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^\top\right)\boldsymbol{y}\| \leq -2\sigma\underline{m}^2\|\boldsymbol{e}\|^2 \tag{5}$$

$$\|\boldsymbol{e}^\top\left(\tfrac{1}{T_s}+2\alpha+2\mathbb{A}^\top\right)\boldsymbol{y}\| \leq \left(\tfrac{1}{T_s}+2\left(\alpha+\overline{A}\right)\right)\overline{m}\|\boldsymbol{e}\|^2, \tag{6}$$

$$-\|\tfrac{\boldsymbol{e}^\top\boldsymbol{M}_{\mathrm{pre}}\boldsymbol{e}}{T_s}\| \leq -\tfrac{1}{T_s}\underline{m}\|\boldsymbol{e}\|^2. \tag{7}$$

By substituting the above inequalities into (1), we have

$$\left(-2\sigma\underline{m}^2+\left(\tfrac{1}{T_s}+2\left(\alpha+\overline{A}\right)\right)\overline{m}-\tfrac{1}{T_s}\underline{m}\right)\|\boldsymbol{e}\|^2 \leq 0. \tag{8}$$

Additionally, invoking (4), the above inequality can be further rearranged as follows:

$$-2\sigma\tfrac{\mu^2}{\chi^2}+\left(\tfrac{1}{T_s}+2\left(\alpha+\overline{A}\right)\right)\mu\chi-\tfrac{1}{T_s}\tfrac{\mu}{\chi} \leq 0. \tag{9}$$

Finally we have the following sufficient condition for the full contraction condition (**??**):

$$\mu \geq \tfrac{1}{2\sigma}\left(2(\alpha+\overline{A})\chi^3+\tfrac{1}{T_s}\left(\chi^3-\chi\right)\right). \tag{10}$$

The right-hand side of (10) is monotonically increasing with respect to $\chi$ for $\chi \geq 1$. Thus, if $\mu$ is sufficiently large and $\chi$ is sufficiently small such that the inequality (10) holds, the full contraction condition (**??**) is guaranteed.

□

### B. Vector-Space Optimization Framework

According to Theorem 1, the following optimization problem can be formulated, as follows:

$$\min_{\boldsymbol{y},s} \quad \|\boldsymbol{y}\|\|\boldsymbol{e}\|-\boldsymbol{y}^\top\boldsymbol{e}+\lambda s \tag{11a}$$

$$\text{subject to} \quad (1), \tag{11b}$$

$$\left(\boldsymbol{e}^\top\boldsymbol{y}-\mu\boldsymbol{e}^\top\boldsymbol{e}\right)^2 \leq (\epsilon+s)\|\boldsymbol{e}\|^2, \tag{11c}$$

$$c(\boldsymbol{y}) \leq 0, \tag{11d}$$

where $s,\epsilon \in \mathbb{R}_{\geq 0}$ denote the slack variable and the allowable small energy variation, respectively of the energy-based constraint (11c) which will be explained later; $\lambda \in \mathbb{R}_{>0}$ denotes the penalty term for the slack variable $s$ and $c(\cdot)$ denotes input constraints, which can be designed to handle the input saturation effect.

The objective function (11a) is designed to minimize the angle between $\boldsymbol{y}$ and $\boldsymbol{e}$, which is equivalent to minimizing the condition number of the contraction metric $\boldsymbol{M}$. Additionally, the last term in (11a) is a penalty term for the slack variable $s$ which will be explained later.

Imposed constraints include the projected contraction constraint (11b), the energy-based constraint (11c), and the input saturation constraint (11d).

The energy-based constraint (11c) is designed to prescribe the error-directional metric scale by enforcing $\boldsymbol{e}^\top\boldsymbol{y} \approx \mu\boldsymbol{e}^\top\boldsymbol{e}$; since $\boldsymbol{y} = \boldsymbol{M}\boldsymbol{e}$, this corresponds to $\boldsymbol{e}^\top\boldsymbol{M}\boldsymbol{e}/\boldsymbol{e}^\top\boldsymbol{e} \approx \mu$. According to Theorem 1, a sufficiently large error-directional metric scale, together with a sufficiently small condition number, promotes satisfaction of the full contraction condition (**??**). However, due to the input saturation constraint (11d), the projected contraction constraint (11b) may conflict with the input saturation constraint (11d), leading to potential infeasibility of the optimization problem (11). To resolve this issue, the slack variable $s$ is introduced to relax the energy-based constraint.

The input saturation constraint (11d) can be designed to directly handle the input saturation effect by defining $c(\boldsymbol{y})$ to represent the admissible input set induced by the saturation function $\mathrm{sat}(\cdot)$.

After the optimization, the contraction metric $\boldsymbol{M}$ can be retrieved while maintaining the condition number and the relationship $\boldsymbol{y} = \boldsymbol{M}\boldsymbol{e}$ as follows:

$$\boldsymbol{M} = \tfrac{1}{\sqrt{\chi}}\,\mathrm{diag}\left(\left(\sqrt{\chi},1,\ldots,1,\tfrac{1}{\sqrt{\chi}}\right)\right)\cdot\tfrac{1}{\|\boldsymbol{e}\|\|\boldsymbol{y}\|}, \tag{12}$$

where $\chi = \frac{1+\sin(\theta)}{1-\sin(\theta)}$ denotes the minimum condition number of $\boldsymbol{M}$ when the angle $\theta$ between $\boldsymbol{y}$ and $\boldsymbol{e}$ is given.

In conclusion, instead of solving the matrix-space optimization problem (**??**), the proposed method solves the vector-space optimization problem (11) to obtain the projected contraction metric $\boldsymbol{y}$. After that, to obtain the contraction metric $\boldsymbol{M}$ for the control law (**??**) and $\boldsymbol{M}_{\mathrm{pre}}$ in (**??**), the contraction metric $\boldsymbol{M}$ is retrieved using (12).

The overall procedure of the proposed method is summarized in Algorithm **??**, in which the optimization problem (11) is solved at each time step.

### C. Illustrative Example in Two-Dimensional Space

To facilitate geometric interpretation, the optimization problem (11) is illustrated in two-dimensional space in Fig. **??**. The projected contraction condition (11b) and the input saturation constraint (11d) restrict the feasible space of $\boldsymbol{y}$ to the outside of the blue line and the inside of the red line, respectively. The green line represents the energy-based constraint (11c), assuming that the allowable energy variation $\epsilon$ is approximately zero.

Without the input saturation constraint, the optimal solution is obtained at point (a) where (11b) and (11c) are active, resulting in the minimum condition number of $\boldsymbol{M}$ and the error-directional metric scale $\mu$. However, when the input saturation constraint is considered, the optimal solution is obtained at point (b) where (11d) and (11c) are active, resulting in relatively large condition number of $\boldsymbol{M}$ and insufficient error-directional metric scale $\mu$. In this case, the slack variable $s$ is increased to relax the energy-based constraint (11c) to satisfy the input saturation constraint, leading to potential violation of sufficiency conditions in Theorem 1.

## V. Validations

## VI. Conclusion

Will be added later.

## Appendix

## References

[1] D. S. Bernstein, *Matrix Mathematics: Theory, Facts, and Formulas (Second Edition)*. Princeton University Press, 2009. [Online]. Available: http://www.jstor.org/stable/j.ctt7t833

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